EXPERIMENT 8

Configuring a Vehicle Service & Repair Request Management Application with Role-Based Access Control (RBAC) in Zoho Creator

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Title

Design and Implementation of a Vehicle Service & Repair Request Management Application with User Permissions, Roles, Role Hierarchy and Data Sharing in Zoho Creator.

Aim

To build a service-request tracking application in Zoho Creator using a structured 13-field service form (Service_Request_Form) and report (Service_Request_Report), and to implement granular role-based access control through a three-level role hierarchy and a custom Read Only data-sharing rule.

Objectives

- To design Service_Request_Form with 13 Basic Fields grouped into four layout panels: Identity, Contact, Vehicle Details and Scheduling.
- To create Service_Request_Report and a dashboard showing service requests, bay capacity utilised and status-wise and service-type-wise workload.
- To define the role hierarchy Workshop Manager → Service Advisor → Front Desk Executive.
- To configure a data-sharing rule from Service Advisor to Front Desk Executive with Read Only access.
- To understand how RBAC separates data ownership from data visibility.

Software / Tools Required

- Zoho Creator (web-based low-code application development platform)
- Vehicle Service & Repair Request Management application created in Zoho Creator
- Web browser with an active internet connection
- Zoho account with access to the application in Edit mode

Introduction / Theory

User Management in Zoho Creator

User management decides who can access an application and what each person may see and do within it. Users are added to an application and associated with permission sets (profiles) and roles; together these control access to forms, reports, pages and records. In the Settings area, the User Permissions section groups these controls under three tabs: Permissions, Roles and Data Sharing.

Role-Based Access Control (RBAC)

RBAC is an access-control model in which rights are assigned to roles instead of individual users, and users acquire rights through the roles they hold. It simplifies administration, supports the principle of least privilege, improves data security and keeps the access policy consistent and easy to audit.

Key Concepts

Concept	Description
Users	Individuals added to the application who sign in to work with its forms, reports and data.

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Concept	Description
Permissions	Named permission sets (profiles) defining what a user may do with components. This application lists *Workshop_Admin_Access*, *Read* and *Write*.
Roles	Positions in the organisational structure (Workshop Manager, Service Advisor, Front Desk Executive) that determine a user's place in the hierarchy and the visibility of records.
Role Hierarchy	A tree in which each role reports to another. A higher role can generally view the records of roles beneath it; the reverse does not apply.
Data Sharing	Share rules that extend record visibility beyond the hierarchy, from one role or user to another, with a defined level of access (Read Only or Read/Write).

Application Specifications

Parameter	Detail
Application Name	Vehicle Service & Repair Request Management
Form Name	Service_Request_Form
Report Name	Service_Request_Report
Form Architecture	13 Basic Fields in 4 layout panels (Identity, Contact, Vehicle Details, Scheduling)
Role Hierarchy	Workshop Manager → Service Advisor → Front Desk Executive
Custom Sharing Rule	Service Advisor → Front Desk Executive (Read Only)

Procedure

1. Sign in to Zoho Creator. The home page lists the available applications. Locate the **Vehicle Service & Repair Request Management** application (Figure 8.1).

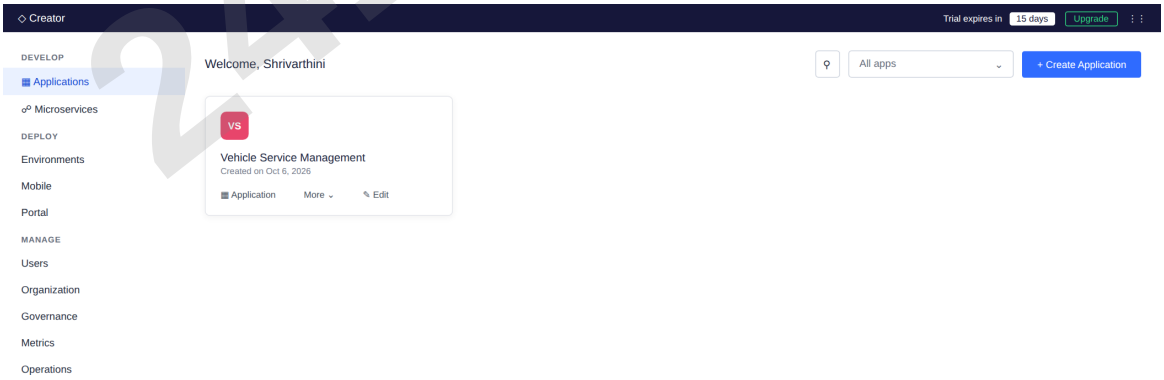


Figure 8.1: Zoho Creator Home – Vehicle Service & Repair Request Management Application

2. Open the application to view its dashboard, which contains the **All Service Requests** report, the *Total Service Requests* and *Bay Capacity Utilised* tiles and the service charts (Figure 8.2).

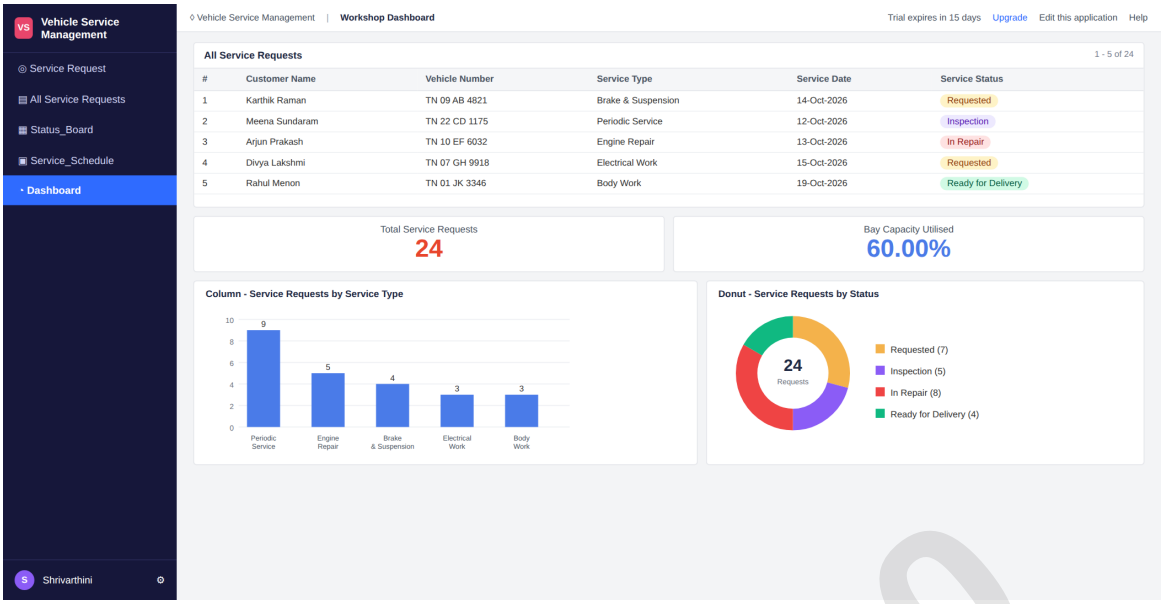


Figure 8.2: Vehicle Service & Repair Request Management Application – Access Mode

3. Enter Edit mode by selecting **Edit** on the application card (Figure 8.1) or **Edit this application** at the top-right (Figure 8.2).

4. Select the **Settings** tab from the top navigation bar (Design, Workflow, Settings).

5. Under the PERMISSIONS group, select **User Permissions**. The Users page is displayed with the tabs Permissions, Roles and Data Sharing.

6. Study the **Permissions** tab. Three permission sets are listed: *Workshop_Admin_Access*, *Read* and *Write*. An **Add Permission** button is also provided (Figure 8.3).

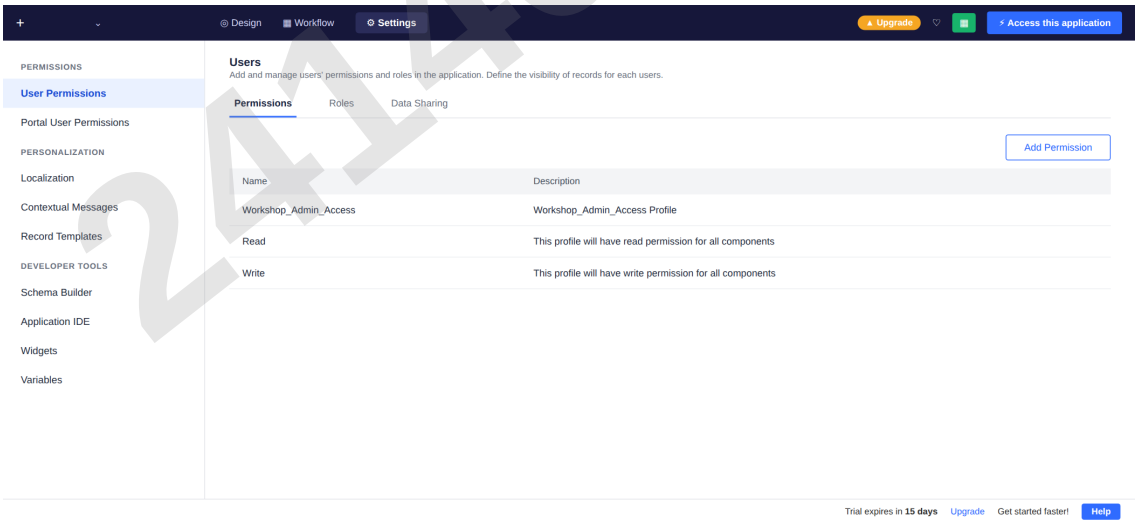


Figure 8.3: User Permissions – Permissions Section

7. Select the **Roles** tab and arrange the role tree as **Workshop Manager → Service Advisor → Front Desk Executive**. The Expand All, Collapse All, Manage Role Hierarchy and Add Role options are available (Figure 8.4).

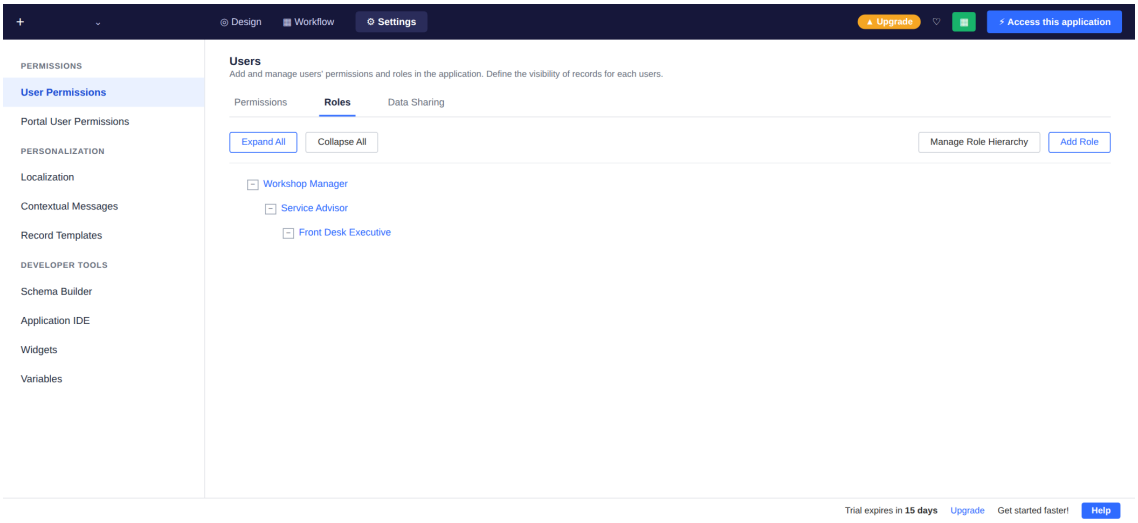


Figure 8.4: Roles and Role Hierarchy

8. Select the **Data Sharing** tab and choose **New Share Rule**. Enter Name: *Service_Request_Form*; Form: *Service Request*; Records shared from: *Roles*. The dropdown offers Roles, Roles and Subordinates, and Users (Figure 8.5).

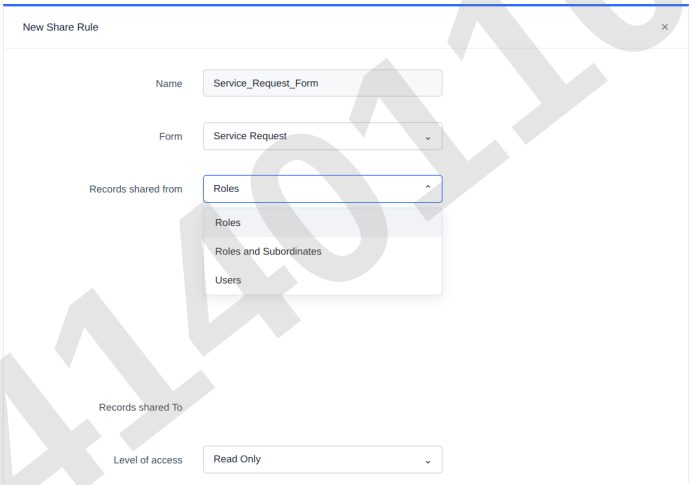


Figure 8.5: New Share Rule – Rule Configuration

9. Set Records shared from → Roles → *Service Advisor*; Records shared To → Roles → *Front Desk Executive*; Level of access → **Read Only**.

10. Select **Save**. The rule appears in the Data Sharing list and can be reopened with Edit Sharing Rule (Figure 8.6).

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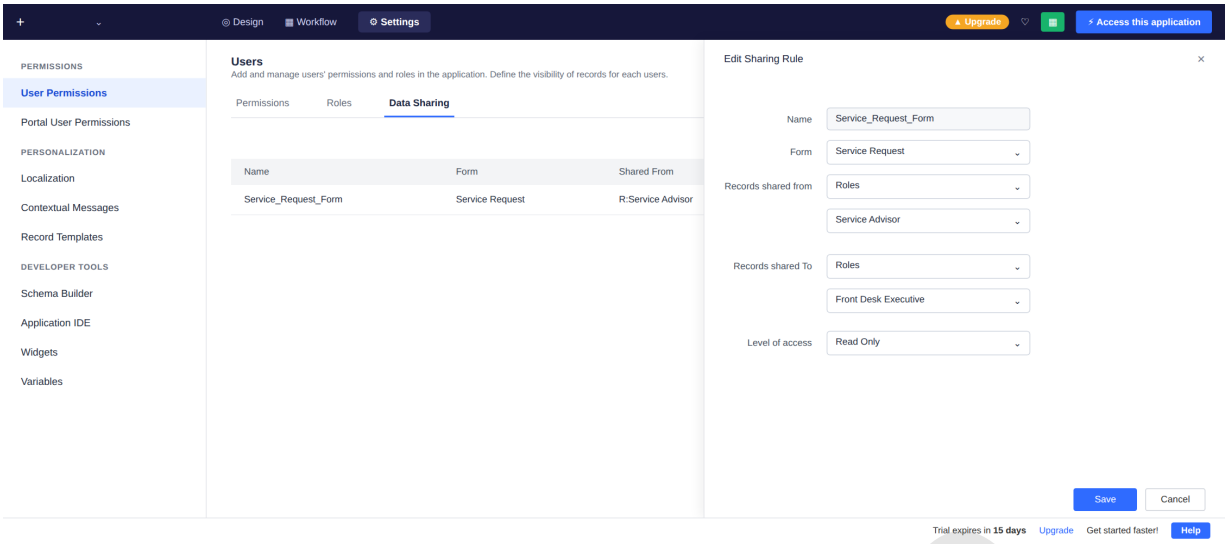


Figure 8.6: Configured Data Sharing Rule (Service Advisor → Front Desk Executive, Read Only)

Data Sharing

The application follows a layered model: **Form** (Service_Request_Form) → **Report** (Service_Request_Report) → **Dashboard** → **Security Layer** (Permissions, Roles, Data Sharing). The role hierarchy runs from the **Workshop Manager** (overall oversight) to the **Service Advisor** (workshop-wide service management) and then the **Front Desk Executive** (day-to-day request intake and scheduling). By default the hierarchy gives only upward visibility: higher roles see the records of the roles below them, but not the reverse.

The custom data-sharing rule shares Service_Request_Form records from the *Service Advisor* role to the *Front Desk Executive* role at **Read Only** level. Read Only preserves data security because Front Desk Executives can view the request details they need for scheduling (customer names, vehicle numbers, service types, slots, contact information) but cannot edit or delete them, so the records held by the Service Advisor remain the single source of truth and cannot be altered downstream, whether by accident or deliberately. It enforces least privilege by granting visibility without modification rights, keeps the scope directional (Advisor to Executive) so access never flows upward or sideways, and leaves a single named rule in the Data Sharing list that documents who can see what, and at what level, making the policy easy to audit.

RBAC Working

Level	Role	Intended Function
1 (Top)	Workshop Manager	Overall authority; able to view the records of roles reporting to it.
2	Service Advisor	Manages workshop-wide service requests; owns the master service request records.
3	Front Desk Executive	Handles request intake and scheduling for the workshop; views shared service requests read-only.

- Users in the Front Desk Executive role work with service request records according to their permission set.
- Because Front Desk Executive reports to the Service Advisor, who reports to the Workshop Manager, higher roles can view the records of the roles below them.
- Lower roles do not automatically gain access to higher-role records; the Read Only share rule grants it in a controlled way.
- The permission sets (Workshop_Admin_Access, Read, Write) further limit the operations each user may carry out.

Result

The following were successfully carried out in the Vehicle Service & Repair Request Management application:

- The Vehicle Service & Repair Request Management application, with Service_Request_Form and Service_Request_Report, was opened and its dashboard viewed.
- The permission sets Workshop_Admin_Access, Read and Write were studied in User Permissions.
- The role hierarchy Workshop Manager → Service Advisor → Front Desk Executive was created.
- A data-sharing rule from Service Advisor to Front Desk Executive with Read Only access was configured and saved.

Conclusion

The experiment demonstrated how permissions, roles, role hierarchy and data sharing together form a role-based access control mechanism in Zoho Creator. Structured service requests with typed fields replace unstructured requests (phone calls, handwritten job cards and chat messages) with clean, consistent and analysable data. The three-level hierarchy and the directional Read Only share rule keep information channels isolated: Front Desk Executives see only what they need and cannot alter master records, while the Workshop Manager and Service Advisor retain oversight. The result is a secure, auditable and scalable vehicle service and repair request management framework.

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